

SOLUTION QUIZ-04 BSE-6A

QUESTION-01

1. **First Address of block:** 172.31.192.0
2. **Total number of addresses:** 8,192
3. **First Useable IP:** 172.31.192.1
4. **Last Useable IP:** 172.31.223.254
5. **Broadcast Address:** 172.31.223.255

QUESTION-02

Total Length (Bytes)	IP Header Length (Bytes)	Identification (ID) (Hexadecimal)	DF Flag	MF Flag	Fragment Offset
596	20	0xABCD	0	1	0
596	20	0xABCD	0	1	72
348	20	0xABCD	0	0	144

SOLUTION QUIZ-04 BSE-6B

QUESTION-01

1. **First Address of block:** 192.168.40.0
2. **Total number of addresses:** 2048
3. **First Useable IP:** 192.168.40.1
4. **Last Useable IP:** 192.168.47.254
5. **Broadcast Address:** 192.168.47.255

QUESTION-02

1.

WAN Side Address (Public)	LAN Side Address (Private)
203.0.113.5:5001	192.168.1.10:3345
203.0.113.5:5002	192.168.1.11:3345

2. Explanation of Distinction When the web server (128.119.245.12) replies, it swaps the source and destination fields from the packet it received.

- **For Host A:** The server sends the reply to 203.0.113.5 on destination port 5001.
- **For Host B:** The server sends the reply to 203.0.113.5 on destination port 5002.

The router examines the **Destination Port** of the incoming packet from the WAN. It looks up this port in the NAT table:

- If the port is **5001**, it rewrites the Destination IP to 192.168.1.10 and Port to 3345.
- If the port is **5002**, it rewrites the Destination IP to 192.168.1.11 and Port to 3345.